

Consolidated simulation plan: latent-class and mixture-model extensions to the carryover and interactions papers

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Objective

Characterise the operating properties of biomarker-treatment interaction inference in aggregated N-of-1 trials when the data-generating process (DGP) departs from the multivariate-normal (MVN) covariance form built into Architecture B, and when the analysis model has access to either the standard `nlme::lme` specification or a mixture-aware alternative. The plan extends the carryover-sensitivity and DGP-comparison studies along two new axes: a latent-class generative axis and a class-aware analysis axis.

Common infrastructure

Shared across all five studies unless stated otherwise.

- **Outcome model.** Fixed analysis formula $Sx \sim bm + t + Dbc + bm:Dbc$, with continuous drug indicator $Dbc = \exp(-\lambda * tsd)$ off-drug and $Dbc = 1$ on-drug. Random intercept $\sim 1|ptID$, residual correlation $corCAR1(form = \sim t|ptID)$. This is the existing `pmsimstats` baseline analysis.
- **Class-aware analysis model.** Joint latent-class linear mixed model via `lcmm::hlme` with two latent classes, biomarker entered in the class-membership submodel, and the same fixed-effects structure for the within-class model. Convergence diagnostics, posterior class entropy, and label-switching checks are stored alongside fitted coefficients.
- **Trial designs.** OL, C0, Hybrid, OL+BDC, generated by `buildtrialdesign()`.
- **Sample sizes.** $N \in \{35, 70, 100\}$, covering the lower end of the trial-relevant range, a midpoint, and an upper anchor for trend extrapolation.
- **Carryover axis.** $t_{1/2,DGP} \in \{0, 0.5, 1.0, 2.0\}$ weeks. Where the analysis model is the MVN linear mixed model, the analysis-side half-life is varied independently across $\{0, 0.5, 1.0, 2.0\}$ to inherit the mis-specification axis from the carryover-sensitivity paper.
- **Replicates.** $n_{reps} = 1000$ per cell for power and bias estimates; $n_{reps} = 5000$ for type I error and class-count recovery cells where rare-event resolution is needed.
- **Performance metrics.** Power for $\beta_{bm:D}$ at $\alpha = 0.05$, type I error under null cells, mean and Monte Carlo standard error of $\hat{\beta}_{bm:D}$, coverage of nominal 95% confidence intervals, convergence rate, fit time. For mixture analyses additionally: entropy of posterior class probabilities, modal class-assignment accuracy, AIC and BIC class-count selection.
- **Reporting standard.** Morris et al. (2019) ADEMP applies to every study. Each study has an a priori ADEMP table giving Aims, Data-generating mechanisms, Estimands, Methods, Performance measures.
- **Common code surface.** A `generateData()` extension is required to produce mixture, combined-moderation, and heterogeneous-random-slopes data; a single dispatch on `dgp_architecture` is preferred over four parallel implementations. The class-aware analysis is wrapped in a `lcmm_analysis()` companion to `lme_analysis()`.

Study 1. Mixture-vs-MVN factorial

Aim. Decompose the gap between Architecture-B power loss under carryover and the corresponding loss under a faithful latent-class DGP into (a) approximation error in the DGP and (b) inefficiency from a mis-specified analysis model.

DGP cells.

1. *Latent-class mixture DGP.* Two-class mixture; class membership $Z_i \sim \text{Bernoulli}(\pi(B_i))$ with logistic gating in B_i ; class-specific drug-response curves of modGompertz form, calibrated so that the marginal $\text{Cor}(B, BR)$ on-drug equals the Architecture-B c_{bm} at the same parameter point.

2. *MVN DGP* (Architecture B as implemented).

Analysis cells.

A. `nlme::lme` baseline.

B. `lcmm::hlme` two-class mixture analysis.

Factorial. A 2×2 DGP-by-analysis grid, crossed with sample size, design, and carryover.

Estimands.

- $\Delta_{\text{recovery}} = \text{power}(\text{mixture DGP, lcmm}) - \text{power}(\text{mixture DGP, lme})$.
- $\Delta_{\text{approx}} = \text{power}(\text{mixture DGP, lme}) - \text{power}(\text{MVN DGP, lme})$.
- Null check: $\text{power}(\text{MVN DGP, lcmm})$ should not exceed $\text{power}(\text{MVN DGP, lme})$; a positive difference would indicate spurious class extraction.

Outputs. A four-panel power-vs-carryover figure (one panel per analysis-DGP cell) and an ADEMP-formatted estimand table reporting both differences with Monte Carlo standard errors.

Dependencies. Mixture DGP code path; `lcmm_analysis()` wrapper.

Study 2. Mean / covariance / combined moderation triptych

Aim. Locate Architecture B within the broader spectrum of biomarker-moderated DGPs and verify the prediction that combined mean-plus-covariance moderation produces carryover-induced power loss intermediate between pure mean and pure covariance moderation.

DGP cells.

1. *Architecture A.* Mean moderation only, as currently implemented.
2. *Architecture B.* Covariance moderation only, as currently implemented.
3. *Combined Armingier-Stein-Wittenberg form.* Simultaneous mean scaling (with coefficient c_{bm}^{mean}) and covariance moderation (with coefficient c_{bm}^{cov}). Calibration anchors: at one calibration point, the on-drug marginal $\text{Cor}(B, BR)$ matches Architectures A and B; the mean and covariance contributions are then varied along a one-dimensional mixing parameter $\alpha \in [0, 1]$ that interpolates from pure mean to pure covariance.

Analysis cell. `nlme::lme` baseline only. Class-aware analysis is not the question here.

Factorial. Three DGP cells (or, for the combined case, five values of α) crossed with sample size, design, and the full carryover and analysis-half-life axes.

Estimands. Power as a function of α at each carryover level. The pre-registered prediction is monotone interpolation: combined-form power loss between A and B at every carryover level.

Outputs. A power surface over $(\alpha, t_{1/2})$ at fixed N and design; sensitivity plots at off-axis N and design.

Dependencies. `generateData()` extension to combined moderation. Calibration of the mean and covariance scales to a common marginal effect-size axis is a non-trivial sub-task and is the first deliverable of this study.

Study 3. Heterogeneous random-slopes DGP

Aim. Test the Verbeke-Lesaffre / Proust-Lima heterogeneous-random-effects representation as a more biologically transparent alternative to Architecture B's covariance-only encoding, while holding the analysis model at the existing `nlme::lme` specification.

DGP cells.

1. *Heterogeneous random slopes.* Two-component mixture over the random-slope distribution on the drug indicator: a responder component with mean slope $\beta_R > 0$ and a non-responder component with mean slope $\beta_{NR} \approx 0$. Biomarker enters the class-membership submodel.
2. *Architecture B baseline,* for direct comparison at matched marginal effect size.

Analysis cells.

A. `nlme::lme` baseline (the audit-chain analysis the paper recommends).

B. `lcmm::hlme` (for completeness; this study borrows the analysis-axis from Study 1 but with the random-slopes DGP rather than the marginal mixture DGP).

Factorial. A 2×2 DGP-by-analysis grid crossed with sample size, design, and carryover.

Estimands. Power loss under carryover for the heterogeneous-random-slopes DGP relative to Architecture B; identifiability metrics for `lcmm` under the random-slopes DGP at small N .

Outputs. Direct power comparison with Architecture B; entropy and class-recovery plots from Study 1 carried over to this DGP.

Dependencies. `generateData()` extension to draw participant-specific random slopes from a two-component mixture. This is a smaller code change than the marginal-mixture DGP in Study 1 and could feasibly precede it.

Study 4. Three-regime breakdown grid

Aim. Map the regions of parameter space where the MVN approximation to a latent-class DGP fails, and quantify the failure.

DGP cells. Three latent-class DGPs, each varying one regime axis with the others fixed at a ‘mild’ baseline.

1. *Class-separation axis.* Vary the drug-response gap Δ between responder and non-responder class-specific Gompertz peaks. At small Δ the marginal response is unimodal; at large Δ it is bimodal. Operationalising metric: the bimodality coefficient of the on-drug BR histogram.
2. *Gating-slope axis.* Vary the logistic slope of $\pi(B_i)$ from gradual (π near 0.5 across the biomarker range) to near-step. A pre-implementation step is to add a thresholded mean-moderation variant of Architecture A, which is the natural comparator at the steep end of this axis.
3. *Class-varying covariance axis.* Vary class-specific residual variance and within-class autocorrelation (responders and non-responders differ in σ and ρ , with class-specific means held equal). This is the latent analogue of `nlme::varIdent`.

Analysis cells. Both `nlme::lme` and `lcmm::hlme`.

Factorial. Three regime axes (run as separate one-dimensional sweeps, not a full three-way factorial), each crossed with sample size and carryover.

Estimands. For each regime, the location on the regime axis at which the MVN-LMM analysis loses more than 5% nominal power relative to the matched mixture analysis; and the location at which the mixture analysis fails identifiability checks (entropy below 0.7 or convergence rate below 90%).

Outputs. Three regime-specific breakdown plots; a synthesis figure marking the joint operating range within which Architecture B is a defensible second-moment summary.

Dependencies. Thresholded mean-moderation variant for the gating-slope axis.

Study 5. Small-N identifiability and type I error

Aim. Establish whether mixture-model analyses are usable at N-of-1-relevant sample sizes, independent of any specific moderation question. This study is a pre-requisite for the substantive interpretation of Studies 1, 3, and the `lcmm` arm of Study 4.

DGP cells.

1. *Single-component, gaussian.* Architecture B with $c_{bm} = 0$. Pure null for the biomarker-treatment interaction.
2. *Single-component, mildly non-gaussian.* Heavy-tailed and mildly skewed residual distributions on the BR factor. Used to replicate the Bauer-Curran (2003) overextraction finding at trial-relevant N .

3. *True two-class*. As in Study 1, at varying class separations.

Analysis cell. `lcmm::hlme` with class counts $K \in \{1, 2, 3\}$.

Estimands.

- Type I error for $\beta_{bm:D}$ under cell 1.
- Class-count selection frequencies (BIC, BLRT) under cells 2 and 3.
- Posterior entropy distribution under cell 3 as a function of N and class separation.
- Convergence rate and mean fit time for each cell.

Outputs. A small- N diagnostic table that tells subsequent studies which cells are interpretable, which require larger N to interpret, and which should be excluded from the substantive power comparisons.

Dependencies. None. This study can run as soon as `lcmm_analysis()` is wrapped.

Sequencing

The dependencies imply the following ordering.

1. **Phase 1: infrastructure.** Implement (a) `lcmm_analysis()` wrapper, (b) heterogeneous-random-slopes DGP path, (c) marginal-mixture DGP path, (d) combined-moderation DGP path, (e) thresholded mean-moderation variant. Items (a) and (b) are independent; (c) and (d) share calibration code with (b); (e) can be built alongside (a).
2. **Phase 2: pre-flight.** Run Study 5. Use its small- N diagnostic table to select interpretable cells in Studies 1, 3, and 4.
3. **Phase 3: substantive studies.** Run Studies 1, 2, 3 in parallel (they share no DGP and no analysis dependencies). Each consumes Phase 1 infrastructure and Phase 2 diagnostics independently.
4. **Phase 4: integration.** Run Study 4. The three regime axes synthesise findings from Studies 1-3 by exploring where the approximations they baseline against begin to break down.

Cross-cutting requirements

- **Pre-flight identifiability.** Every mixture-DGP cell must pass a positive-definiteness and convergence pre-flight analogous to `analysis/scripts/figure4/05-pd-diagnostics.R` before entering the production simulation grid.
- **Calibration to a common effect-size scale.** The biomarker-treatment interaction is parameterised differently across the four DGPs (Architecture A as a regression coefficient, Architecture B as a correlation, mixture DGPs as a class-separation Δ paired with a gating slope, combined form as a mixing parameter α). A single calibration sub-study, performed once and reused across Studies 1-4, expresses each as the marginal Cohen's-style standardised effect size at a reference time and drug state. Without this, cross-study power comparisons are not interpretable.
- **Null cells for every study.** Each study includes a $c_{bm} = 0$ (or $\Delta = 0$) null cell for type I error. The temptation to assume the carryover-sensitivity paper's null calibration carries over should be resisted: each new DGP needs its own null check.
- **Computational budget.** Architecture B at the carryover-sensitivity paper's grid scale runs in approximately $O(\text{designs} \times t_{1/2} \times N \times c_{bm} \times n_{\text{reps}})$. Adding mixture DGPs and `lcmm` analyses inflates per-cell time substantially; a budget estimate from a small pilot at one cell per study, performed during Phase 1, is required before Phase 3 commits.

Deliverables

- A `simulations.R` master driver script under `analysis/scripts/latent-class-mixture-application/` that orchestrates the five studies through the standard `furrr::future_map` and chunked-save pattern.
- Five study-specific summary scripts producing the figures and ADEMP tables described above.
- A `report.Rmd` results section that organises findings by study, with the cross-study calibration sub-study reported up front so that downstream effect-size axes are interpretable on a common scale.

- Pre-registered ADEMP tables for each study, committed before the production runs are executed.

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